

Determinants of Trading Profits of Individual Traders: Risk Premia or Information?

Michaël Dewally
Marquette University

Louis Ederington
University of Oklahoma

Chitru Fernando
University of Oklahoma

March 2010
Initial Draft: August 2009

JEL Classifications: G12, G13

Keywords: risk premium, trading profits, energy futures, speculators, hedgers, market makers

Contact Info: Michaël Dewally, Assistant Professor of Finance, Marquette University, College of Business Administration, Straz Hall, 318, P.O. Box 1881, Milwaukee, WI 53201-1881, Tel: (414) 288-1442, michael.dewally@marquette.edu.

Louis Ederington (corresponding author), Michael F. Price Chair in Finance, University of Oklahoma, Department of Finance, Michael F. Price College of Business, Rm 205. Adams Hall, Norman, OK 73019-4005, Tel: (405) 325-2697, lederington@ou.edu.

Chitru S. Fernando, John and Donnie Brock Chair, Associate Professor of Finance and Director, Energy Solutions Center, Michael F. Price College of Business, University of Oklahoma, Rm. 205, Norman, OK 73019-4005, Tel: (405) 325-2906, cfernando@ou.edu.

We have benefitted from comments on earlier drafts by seminar participants at the University of Oklahoma and the US Commodity Futures Trading Commission, including Pradeep Yadav, Scott Linn, Bill Megginson, James Moser, Jeffery Harris, Michel Robe, and Bahattin Büyüksahin. We have also benefitted from useful discussions with US Energy Information Administration officials. We thank the US Department of Energy for providing the proprietary dis-aggregated energy derivatives trading data used in this study. We are responsible for all remaining errors. Part of this research was completed while Louis Ederington was Visiting Professor at the University of Melbourne, which he thanks for their support.

Determinants of Trading Profits of Individual Traders: Risk Premia or Information?

Abstract

Using a unique proprietary data set of trades by all large traders in the crude oil, gasoline, and heating oil futures markets, we explore determinants of their individual trading profits/losses. Consistent with the risk premium hypothesis, hedgers' mean trading profits are significantly negative while speculators' trading profits (hedge funds especially) are significantly positive. Moreover, the profits of individual traders (whether speculators or hedgers) are a strong positive function of the extent to which the trader shorts (longs) when likely hedgers in the aggregate are long (short). While some individual traders may have an informational advantage, the trading profits of speculators in general, and hedge funds in particular, are due to their employing trading strategies which take advantage of the risk premium, i.e., longing (shorting) when hedgers in the aggregate are net short (long). Thus, our evidence indicates that speculator profits are primarily due to the liquidity and risk absorption services they provide hedgers. Market makers realize significant losses on their overnight holdings, which is consistent with findings in prior studies for other markets that any information advantages they may possess are short-lived. We also find that (excepting households) speculator profitability is a positive function of the rates at which speculators turn over their portfolios.

Determinants of Trading Profits of Individual Traders: Risk Premia or Information?

"Speculation in the food and fuel markets is not illegal, but that does not mean it is not very hurtful. They are artificially inflating the price of food and oil and causing real suffering for millions and millions of people and businesses." Sen. Joe Lieberman as quoted in the *New York Times*, June 28, 2008.

"I'm afraid I don't have much to contribute [to the current debate on commodity speculation] since Milton Friedman convinced me long ago that profitable speculation is stabilizing and destabilizing speculation is unprofitable." Bob McTeer, Former President, Federal Reserve Bank of Dallas, as quoted in the *Wall Street Journal*, July 14, 2009.

1. Introduction

Recent price volatility in commodity markets has renewed interest among academics, practitioners and policymakers in the role and impact of individual traders, especially speculators, in derivatives markets, and reopened debate on the appropriate role of speculation in these markets. While this debate has spawned several studies,¹ one of the most fundamental underlying issues, the extent and determinants of individual traders' trading profits, has not been examined. The degree to which differential trading profits/losses among security traders represent differences in risk taking, differences in information or skill, or luck, is one of the most basic issues in finance. Addressing it in the context of commodity derivatives markets is essential to understanding the incentives, behavior, and economic roles of various trader types. This paper seeks to provide evidence on this issue by examining determinants of the trading profits of individual traders in three energy futures markets: crude oil, gasoline, and heating oil.

According to the traditional risk premium (or normal backwardation) theory of Keynes (1930) and Hicks (1939), expected futures trading profits should be positive for speculators and negative for hedgers. The presumption is that rational speculators will not enter the market unless

¹ See, for example, Haigh, Hranaiova, and Overdahl (2007), Interagency Task Force on Commodity Markets (2008), Büyükşahin, Haigh, Harris, Overdahl, and Robe (2008), and Büyükşahin and Harris (2009).

expected profits from speculation are positive. Since futures market trading is a zero-sum game before transaction costs, this means that hedgers lose on their futures positions on average. If most hedgers have long positions in the underlying asset (as Keynes and Hicks assumed) so hedge by shorting in the futures market, futures prices will be pushed below expected future spot prices (backwardation) and speculators can make positive profits on average by going long. If most hedgers are short in the underlying asset and long in the futures market, futures prices will be biased upward (contango) and speculators can make expected profits by shorting futures.

Heretofore, this theory has primarily been tested by relating ex post returns on hypothetical futures positions to aggregate open interest positions of commercial and non-commercial traders as reported in the Commodity Futures Trading Commission (CFTC)'s weekly "Commitments of Traders" (COT) report. Presuming that commercial traders are more likely to be hedgers than non-commercial traders,² the risk premium hypothesis predicts positive average returns on long (short) positions when commercial traders in the aggregate are net short (long). Using this approach, Bessembinder (1992), Bessembinder and Chan (1992), De Roon et al. (2000), and Wang (2001) find evidence supporting the risk premium theory. Kolb (1992) does not. Fama and French (1987) find some evidence for a risk premium but conclude it is too weak "to resolve the long-standing controversy about the existence of non-zero expected premiums." (1987, p. 72).

The luck hypothesis and evidence is most forcefully stated by Hartzmark (1987, 1991) in two of the very few papers that present evidence on actual trades. Hartzmark (1987) finds that commercial traders earn small positive futures trading profits on average while non-commercial trading profits are negative - directly the opposite of the risk premium hypothesis prediction. In a follow-up (Hartzmark, 1991), he finds: 1) that the number of traders with consistent forecasting ability (as measured by consistent profits) is no more than one would expect due to chance, and 2) that winners in the first half of his study period are no more likely than first half losers to be winners in the second half. He concludes that "luck determines trader performance."

² This presumption has been questioned by Ederington and Lee (2002).

A growing body of evidence outside of futures markets - stock markets in particular - calls both the risk premium and luck hypotheses into question. First, a number of studies, including Odean (1998a, 1998b, 1999), and Barber and Odean (2000), argue that much individual investor trading is influenced by behavioral biases, such as the disposition effect and overconfidence, so that individual investors, particularly those who trade frequently, earn below average profits. Applied to futures markets, this would suggest that some speculators may not be deterred by losses to the extent predicted by a model of rational decision making. In other words, while the expected profits from speculation might be negative if rationally calculated, some speculators may overestimate the likelihood of positive outcomes due to overconfidence.

Second, a number of stock market studies find that informational and/or skill levels differ between investors leading to differential trading profits - at least before transaction costs. Papers by Carhart (1997), Grinblat and Titman (1989, 1993), Wermers (2000), and Chan et al. (2000) among others find that some mutual funds have superior stock picking ability while Ackerman et al. (1999), Ibbotson and Chen (2005), Kosowski et al. (2007), and Jagannathan et al. (2008) find the same for hedge funds. We would expect informational differences to be at least as great in the energy futures markets which we examine as in the stock market - particularly since companies are not under the equal disclosure requirements in these markets that they are in stock markets. Large energy producers may be able to forecast supply and demand from their own production, inventories, and order flow while pipelines and marketers seem particularly well placed to observe flows, orders, and inventories. Hedge funds may possess unique trading skills. Consistent with the differential information hypothesis, Leuthold et al. (1994) find in the pork bellies futures market that “a subset of elite traders possesses significant forecasting ability,” leading to consistent trading profits.

Third, the distinction between hedgers and speculators is not sharp since many hedging programs apparently include some speculation. Dolde (1993), Bodnar, Hayt and Marston (1998) and Glaum (2002) provide survey evidence suggesting that the practice of selective hedging, i.e.,

firms incorporating their views about future market movements into their hedging programs, is widespread in the U.S. and Europe. Stulz (1996) suggests that selective hedging would be rational if firms possess the ability to forecast future market movements based on private information they acquire in the course of their business. However, Adam and Fernando (2006) and Brown, Crabb, and Haushalter (2006) find no evidence that firms are able to outperform the market by selective hedging.

In this paper, we test how trader profits in three large energy futures markets: crude oil, gasoline, and heating oil depend on: 1) whether the trader is most likely a hedger or speculator, 2) possible informational and skill differences, 3) the trading strategy employed, and/or 4) luck, using a unique proprietary data set of trades by individual large and mid-size traders over 46 months. These markets offer a particularly apt opportunity to study these issues since recent high volatility in energy prices has been a primary driver of interest in this topic among academics, practitioners and policymakers.

As part of its market surveillance, the CFTC requires brokers to report daily open interest positions of all traders whose open interest positions exceed thresholds set to capture about 70% - 90% of total open interest. While normally available to researchers only in the highly aggregated form of the Commitments of Traders" (COT) report used in previous studies, the Department of Energy (DOE) provided us with the disaggregated data on the open interest positions of individual reporting traders in the NYMEX's crude oil, heating oil and gasoline futures markets. We utilize this disaggregate daily data on individual traders to calculate trader profits. Since both the likelihood that a trader is primarily hedging or speculating and their access to information should differ by industry or line of business, each trader is classified into one of the eleven trader categories described in Table 1: refiners, independent producers, marketers/distributors/pipelines, large energy consumers, commercial banks, energy traders, hedge funds, households, investment banks, market makers, and unknown (unclassified). We also hypothesize that trading profits will

differ with the trader's size, experience, and trading strategy. Contradicting Hartzmark's (1987, 1991) luck hypothesis, we document persistent profit differences among individual traders.

We report a series of findings in support of the risk premium hypothesis: (1) we find a significant relation between the risk premium and net hedging pressure, i.e., that net longing (shorting) by hedgers pushes futures prices above (below) expected future spot prices, creating potential profit opportunities for speculators; (2) likely hedgers, as identified by their line of business, tend to have trading losses while likely speculators, hedge funds in particular, turn consistent profits; (3) individual trader profits are a positive function of the extent to which they take positions opposite to the aggregate positions of likely hedgers, i.e., longing (shorting) when likely hedgers in the aggregate are net short (long); (4) individual trader profits are higher if they tend to long when markets are in backwardation and short when in contango; (5) the profits of hedge funds, which are the highest among our eleven trader types, are almost totally due to their exploitation of the risk premium, i.e., longing when likely hedgers in the aggregate are net short and shorting when likely hedgers are net long.

While we are unable to measure their earnings from intra-day trading, we find that floor traders (the major market makers in these markets in our data period) experience sizable trading losses from their overnight holdings, which is consistent with prior studies showing that any information advantages possessed by market makers are short-lived at best (Silber, 1984; Hasbrouck and Sofianos, 1993; Yao, 1997). Excepting households, trading profits tend to be higher for more active speculators. However there is no evidence that larger traders have an information advantage. Consistent with the evidence in Büyüksahin et al. (2008), there is weak evidence that spread traders were able to profit from mispricings during this period.

While it is possible that some individual traders have superior information or skill, there is no evidence that some trader types, such as hedge funds, have an informational advantage since we find that the mean profit differences among our eleven trader types (as well as between likely hedgers and speculators) are completely explained by the extent to which individual traders of each

type exploit the risk premium by longing (shorting) when likely hedgers in the aggregate are short (long). Thus, our evidence indicates that the positive trading profits of speculators in general, and hedge funds in particular, are due to the risk transfer and liquidity services they provide hedgers - not superior information.

The remainder of the paper is organized as follows. Our unique data set is described in the next section and our profit measures in section 3. In section 4, we develop our hypotheses and independent variables. Results are described in section 5. Section 6 concludes the paper.

2. Data

Our data comes from the CFTC's Large Trader Reporting System (LTRS) which is described by Haigh et al. (2007). The CFTC requires daily reports on the holdings of all traders whose total open interest position exceeds set thresholds, currently 350 contracts for crude oil, 250 for heating oil, and 150 for gasoline. According to the CFTC, these thresholds are set so that the reported positions normally account for 70% to 90% of total open interest (CFTC website, August 2008). The remaining 10% to 30% represents unreported positions of smaller traders. From this data, the CFTC compiles the weekly COT report on the aggregate open interest positions of commercial and non-commercial traders used in past tests of the risk premium hypothesis by Bessembinder (1992), Bessembinder and Chan (1992), De Roon et al. (2000), Wang (2001) and others. While normally available only in this highly aggregated form, the Department of Energy (DOE) provided us with the disaggregated data on the open interest positions of individual reporting traders in the NYMEX's crude oil, heating oil (also known as #2 fuel oil), and gasoline futures markets from June 1993 through March 1997.³ On average over this period, the traders in our data set accounted for 77.7% of open interest in the crude oil futures market, 81.3% of gasoline open interest, and 68.4% of heating oil open interest.

³ This data which was provided to us in a form which concealed the identities of individual traders is no longer available to researchers even in this anonymous form.

To classify traders as commercial or non-commercial, the CFTC asks on the reporting form: “Is the reporting trader engaged in business activities hedged by the use of futures and options markets?” If the answer is yes, the trader is classified as “commercial” and if no as “noncommercial.” Since the traders self-select (though the CFTC staff may overrule them), there may be a bias toward the hedger category. Note that the question refers to “markets,” not “*this* market?” Ederington and Lee (2002) argue that in the energy futures markets the “commercial” category includes many likely speculators.

The CFTC also assigns sub-classifications, which are described in Haigh et al. (2007) and Büyükşahin et al. (2008), to most traders. We utilize the CFTC’s non-commercial sub-classifications which include: commodity pool operators, commodity trading advisors, managed money, futures commission merchants, floor traders, and floor brokers. According to Haigh et al. (2007), the first three sub-classifications are similar and composed primarily of hedge funds so, like Haigh et al. (2007), we combine these three into a single hedge fund/managed money category. Because trades by floor brokers and futures commission merchants for their clients are reported for the latter, the trades reported under these two sub-classifications represent (like floor traders) trades for their own account so we combine them with floor traders. As described by Silber (1984) and Bryant and Haigh (2004), floor traders were the major market makers in futures markets during our time period. Non-commercial traders with no subclassification code are classified as households (individuals). In summary, we assign the CFTC’s non-commercial traders to three trader types: hedge funds, market makers/floor traders, and households.

The CFTC’s sub-classifications of commercial traders, e.g., manufacturer, dealer/merchant, etc., proved less useful so, with the assistance of the DOE, we assigned the subclassifications described in Table 1: refiners, independent producers, marketers/distributors/pipelines (MDPs), large consumers, commercial banks, investment banks and brokers, and energy traders.⁴ The latter

⁴ This line-of-business classification was done in a manner which preserved trader anonymity from the authors. The CFTC data set identifies traders by number only. Separately, The Office of Policy at the U.S. Department of Energy provided us with the names (without

are primarily offshore hedge funds. We separate energy producers into those with refining capacity (refiners) and without (independent producers) since the latter are clearly long in the physical crude oil market while the former could be either long or short.⁵ A residual group of smaller commercial traders could not be classified.

In addition to the eleven trader types listed in Table 1, we separate the traders into: 1) likely hedgers, 2) likely speculators, 3) market makers, and 4) others. A limitation of our study is that we know only the traders' futures market positions. We can not observe their cash market, forward, or swap positions.⁶ Hence, our judgment of the likelihood that a trader is hedging or speculating is based partially on their line-of-business.⁷ We regard the first five of the commercial classifications listed in Table 1 as the traders most likely to be hedging. The first four classifications: (1) refiners, (2) energy independents, (3) MDPs, and (4) large consumers all have sizable cash and/or forward energy market positions which they could be hedging and banking regulations restrict commercial banks (the fifth classification) to hedging activities. Conversely, we view hedge funds and individuals from the CFTC's non-commercial category and energy traders from the commercial category as likely speculators since they have no known physical or forward market positions. Investment banks/brokers are the major unknown on the hedging/speculating spectrum since they

numbers attached) of the traders in our data set. Commercial and investment banks/brokers were identified by two-digit SIC codes. With assistance from the Office of Policy, the remaining traders were classified by line-of-business. The Office of Policy then matched the line-of-business codes, but not names, to each trader number and provided the data to us.

⁵ Ederington and Lee (2002) regard independent producers as likely speculators in the heating oil market since they do not produce heating oil. Since they produce crude oil, we regard them as potential hedgers in that market. To avoid complicating the results presentation we keep the classifications the same across all three markets.

⁶ We do have some data on their option positions but in an aggregated form which precludes any trading profit calculations. Fortunately, it is clear from this data that, for almost all traders, option positions were quite small relative to their futures positions during this period.

⁷ Below we will present some evidence consistent with these classifications.

could either be hedging their OTC swap positions or speculating. Hence, we consider them separately. Market makers/floor traders are also treated separately.

As noted above, the LTRS data which we utilize is normally available only in highly aggregated form. Like us, Hartzmark (1987, 1991) and Leuthold et al. (1994) use dis-aggregated LTRS data to calculate profits of individual traders. but they only use the commercial and non-commercial classifications. Haigh et al. (2007), Büyüksahin et al. (2008), and Büyüksahin and Harris (2009) utilize the CFTC subclassifications but do not calculate individual trader profits.

Our data set contains 939 traders for which there are 1,059,616 trader/day/contract observations and 486,334 trades. However, many of these 939 traders were only active a few days during our data period or made small trades. To ensure continuous series for most traders,⁸ and to make the trader classification task described above more manageable, for inclusion in our final sample we require that the trader holds an open interest position at least 100 days during our period and make at least 50 trades. This results in a final sample of 382 traders. Since they account for 96.3% of the trader/day/contract observations and 97.9% of total open interest, we lose very little by restricting the sample to these 382 more active traders. Descriptive statistics are reported in Table 2 for our three markets.⁹ Note that the crude oil market is the largest – accounting for 64.9% of open interest in terms of contracts and 60.6% in dollar terms, and 50.6%, of our observations. Times-to-expiration tend to be longer in the crude oil market.

Descriptive statistics for our eleven trader types are in Table 3. As shown there, in terms of both open interest and trades, the energy futures markets are dominated by refiners, investment banks, and MDPs which together account for 68.7% of open interest and 57.7% of trades. Major

⁸ Brokers are only required to report a trader's open interest if it exceeds the set threshold. In practice, once a trading system is initiated for a trader it appears to normally continue when open interest falls below the minimum. Nonetheless, reporting could stop once a trader's open interest falls below the minimum so this requirement increases the likelihood that our observations for a trader are complete.

⁹ The gross open interest reported in Table 1 = total long positions + total short positions. This differs from the usual open interest figure reported for the market as a whole – total long or short positions.

energy consumers, such as airlines, trucking companies, and utilities, are much less active. Combined, the similar hedge fund and energy trader classifications represent roughly eleven percent of both open interest and trades.¹⁰ Making numerous small trades, market makers account for 12.9% of trades but only 4.4% of open interest. These figures undoubtedly understate market maker trading since we only capture positions held overnight. While numerous, the unclassified traders represent a minor portion of trading and open interest.

The figures in Table 3 indicate some speculation or selective hedging by energy firms. To hedge future sales, refiners should short gasoline and heating oil contracts.¹¹ Yet, long positions account for roughly a third of their open interest in these contracts. While independent producers would hedge their physical positions by shorting short crude oil, about 30% of their positions are long. The consumer category consists primarily of airlines which one would expect to hedge their fuel needs by longing fuel oil contracts. That is the case for the vast majority (85% of open interest) but not all. Commercial bank open interest is concentrated in crude oil where they are generally short - the expected pattern if hedging loans to energy producers. Households are generally long.

Prices of the nearby contracts over our data period are graphed in Figure 1. For comparison with crude oil prices, which are quoted per barrel, gasoline and heating oil prices are converted from gallons to barrels. As shown in Figure 1, prices both rose and fell over our data period. To illustrate periods of backwardation and contango, in Figure 2 we graph ratios of the price of the third month contract divided by the price of the nearby contract. Periods of both backwardation and contango are observed.

¹⁰ According to US Senate Staff Report (2006), Haigh et al. (2007) and Büyüksahin et al. (2008) hedge fund trading in these markets increased substantially subsequent to our data period - perhaps attracted by the hedge fund profits which we document below.

¹¹ Whether refiners hedge by longing or shorting crude oil futures would depend on whether their crude production exceeds their refining capacity or the reverse.

3. Profit measures

As explained above, the LTRS data report end-of-day open interest positions of all large traders. If a trader's open interest in a particular futures contract changes between the close of day $t-1$ and close of day t , clearly a trade occurred on day t . However, we miss any trades reversed the same day. Our profit measure is the daily holding period profit resulting from marking the traders' positions to market at the end of each day. If there is no trade on day t , the profit/loss on trader i 's holding in a given futures contract is $X_{i,c,m,t-1}(P_{c,m,t}-P_{c,m,t-1})$ where $X_{i,c,m,t}$ is trader i 's open interest position in contract type c (crude oil, gasoline, or heating oil) maturing in month m at the close on day t and $P_{c,m,t}$ is the closing price of contract $c-m$ (henceforth we will use the term “ $c-m$ ” to designate a contract of type c expiring in month m) on day t . If i 's position in contract $c-m$ is short on day t , $X_{i,c,m,t} < 0$. Suppose trader i longs contract $c-m$ on day t so $X_{i,c,m,t} > X_{i,c,m,t-1}$. Since we don't observe actual trade prices, we approximate using closing prices. For the day t purchase price, we use an average of the closing prices on days t and $t-1$ so the holding period profit/loss on day t is calculated as $X_{i,c,m,t-1}(P_{c,m,t}-P_{c,m,t-1}) + (X_{i,c,m,t}-X_{i,c,m,t-1})[.5(P_{c,m,t}+P_{c,m,t-1})]$. If i shorts contract $c-m$ on day t , the profit loss is $X_{i,c,m,t}(P_{c,m,t}-P_{c,m,t-1}) + (X_{i,c,m,t-1}-X_{i,c,m,t})[.5(P_{c,m,t}+P_{c,m,t-1})-P_{c,m,t-1}]$. So whether i longs, shorts or does not trade, her profit/loss on contract $c-m$ on day t may be calculated as:

$$PL_{i,c,m,t} = .5(X_{i,c,m,t} + X_{i,c,m,t-1})(P_{c,m,t} - P_{c,m,t-1}) \quad (1)$$

Each day we sum trader i 's profits/losses over all contracts obtaining:

$$PL_{i,t} = \sum_{c=1}^3 \sum_{m=1}^M PL_{i,c,m,t} \quad (2)$$

Note that, except for our use of estimated day t trade prices, this profit/loss measure is identical to the mark-to-market calculation used to credit profits and debit losses to a trader's account and to determine margin calls. Tests of the risk premium hypothesis by Bessembinder and Chan (1992), De Roon et al. (2000), and Wang (2001) and others also use closing prices but for hypothetical trades since they do not observe open interest positions of actual traders. Since $PL_{i,t}$ does not

incorporate bid/ask spreads or other transaction costs, trading profits after trading costs are slightly overstated for all traders except market makers for whom they are understated.

Since dollar profits and losses vary with the size of a trader's position, we also calculate percentage profits. Since futures involve no up-front investment, returns do not have the same meaning as in other markets. Moreover traders may hold long, short or spread positions. Nonetheless, return-like measures are useful so we calculate trader i 's profit on contract c - m on day t as a percentage of her average dollar open interest in that contract, specifically:

$$\%PL_{i,c,m,t} = \frac{PL_{i,c,m,t}}{.5[|X_{i,c,m,t}|P_{c,m,t} + |X_{i,c,m,t-1}|P_{c,m,t-1}]} * 100 \quad (3)$$

Since trader i may hold positions in numerous contracts, her percentage profit on day t is calculated by weighting the percentage profit/loss on each contract c - m by its dollar open interest as a percent of the total dollar open interest across all contracts, i.e.,

$$\%PL_{i,t} = \frac{\sum_{c=1}^3 \sum_{m=1}^M \%PL_{i,c,m,t} \cdot .5[|X_{i,c,m,t}|P_{c,m,t} + |X_{i,c,m,t-1}|P_{c,m,t-1}]}{\sum_{c=1}^3 \sum_{m=1}^M .5[|X_{i,c,m,t}|P_{c,m,t} + |X_{i,c,m,t-1}|P_{c,m,t-1}]} \quad (4)$$

For example, if a trader longs one contract at \$100 and shorts it 50 days later at \$100 but the price fluctuates up and down in the interim, we calculate 50 separate non-zero daily mark-to-market profit/loss figures instead of one round trip trade with zero net profit.

Finally, trader i 's average daily percentage profit/loss over the entire period is calculated by weighting each day's $\%PL_{i,t}$ by the size of her average open interest position on day t :

$$\%PL_i = \frac{\sum_{t=1}^T \sum_{c=1}^3 \sum_{m=1}^M \%PL_{i,c,m,t} \cdot .5[|X_{i,c,m,t}|P_{c,m,t} + |X_{i,c,m,t-1}|P_{c,m,t-1}]}{\sum_{t=1}^T \sum_{c=1}^3 \sum_{m=1}^M .5[|X_{i,c,m,t}|P_{c,m,t} + |X_{i,c,m,t-1}|P_{c,m,t-1}]} \quad (5)$$

There are several suspicious observations in our data which could not be checked. For instance, there are a few instances in which a position (with an identical number of contracts

throughout) is recorded as short for several days, suddenly switches to long for a day or two, and then back to short. To guard against outliers caused by possible data errors contaminating our results, we winsorize by replacing $\%PL_{i,t}$ values in the lower 1% tail of i 's daily profit figures with i 's 1% value and similarly for values in the upper 1% tail.

4. Hypothesized profit determinants

In contrast to Hartzmark's (1987, 1991) luck hypothesis, we expect differences among traders in either information, skill, trading objectives (i.e., hedging versus speculation), and/or trading strategies to result in consistent profits by some traders and losses by others. There are two underlying theoretical hypotheses on which we seek to present evidence. The first is the risk premium hypothesis that expected trading profits must be positive to entice speculators into the market which (since trading is a zero sum game prior to transaction costs) implies negative expected trading profits for hedgers. The second underlying hypothesis is that more informed or skilled traders will make profits at the expense of less informed – at least before deducting information costs. Since we cannot observe directly who is hedging, who is speculating, and who is more informed, we relate profitability to trader and trade characteristics as outlined below.

4.1. Trader type

As argued above, we expect a high proportion of hedgers in the first five categories listed in Table 1: refiners, independent producers, MDPs, large consumers, and commercial banks - the first four because of their spot or forward market positions and commercial banks because of regulations. Conversely, we expect the next three: energy traders, hedge funds, and households to be dominated by speculators. Therefore, according to the risk premium hypothesis, trading profits should be negative on average for refiners, independent producers, MDPs, large consumers, and commercial banks, and positive for energy traders, hedge funds, and households. It is quite possible that information and/or skill levels also differ among trader types leading to profit differences.

MDPs might be more informed about physical market conditions and investment banks and hedge funds better informed about trading strategies. Determining whether any profit differences among trader types reflect hedging versus speculation or information/skill differences will be a major task of the paper.

Market makers are a separate and interesting case. A basic tenet of market microstructure theory is that market makers tend to lose on trades with more informed investors (Bagehot, 1971; Glosten and Milgrom, 1985). In other words, informed traders tend to buy (sell) and market makers therefore tend to sell (buy) before a price increase (decrease). The bid-ask spread compensates for these asymmetric information losses. Since our measure of trading profits based on closing prices does not include their bid/ask spread profits, this theory would imply that our trading profit measure should be negative on average for market makers. On the other hand, it has been argued, e.g., Manaster and Mann (1996), Brown and Zhang (1997) and Ready (1999), that due to their presence on the floor where they can observe the order flow, floor traders (the main futures market makers during our data period) have an information advantage over most other traders, implying positive average profits. Since we only observe positions held overnight, not market makers' more numerous intra-day trades, this latter argument may not apply in our case. Nonetheless, our data provides a unique opportunity to explore market maker profits in commodity futures markets - at least on positions they hold overnight.

4.2. Long or short relative to hedgers in general

Turning from trader groups to individual traders, we expect their profits to depend on their trading characteristics. As we would interpret the risk premium hypothesis, a trader's expected profits should depend not just on whether she is a hedger or speculator but on her long/short position relative to the majority of hedgers. Consider for instance the position of an airline which hedges its future fuel purchases by longing fuel oil (heating oil) futures - which, as documented in Table 3, is what they normally do. In the fuel oil market, energy consumers are grossly

outnumbered by refiners and MDPs whom one would expect to normally be hedging future sales by shorting heating oil futures. If their shorting activity pushes futures prices below expected future spot prices, the airlines' expected trading profits should be positive - like those of speculators. The airline hedger might be willing to accept expected losses on average but does not have to. Hence, an individual trader's expected profits should be negative when her long/short position matches that of hedgers in general and positive when it differs.

Therefore, we hypothesize that an individual trader's trading profits will tend to be negative when the sign of her open interest position (long or short) in a specific contract matches the sign (long or short) of the aggregate open interest position of the five trader categories we classify as likely hedgers in Table 1 and positive when her long/short position differs from theirs. To test this, for each of our 382 traders, we define a hedger concordance (HC) measure of the extent to which the signs of trader i 's open interest positions match the signs of the aggregate positions of likely hedgers. Specifically,

$$HC_i = \frac{\sum_{c=1}^C \sum_{m=1}^M \sum_{t=1}^T D_{i,c,m,t} |X_{i,c,m,t}|}{\sum_{c=1}^C \sum_{m=1}^M \sum_{t=1}^T |X_{i,c,m,t}|} \quad (6)$$

where $X_{i,c,m,t}$ is trader i 's open interest position (long or short) in contract c - m on day t . $D_{i,c,m,t} = 1$ if $X_{i,c,m,t}$ is long when likely hedgers are net long in the aggregate or short when likely hedgers are net short in the aggregate. $D_{i,c,m,t} = 0$ if $X_{i,c,m,t}$ is long when likely hedgers are net short in the aggregate or short when likely hedgers are net long in the aggregate. So HC_i represents the percentage of times the sign of trader i 's position matches that of likely hedgers in the aggregate weighted by the size of the position, and varies between zero and one. Of the 46,253 contract/maturity/day observations in our sample, likely hedgers are net short in 58.7% and net long in 41.3%. We hypothesize that trader i 's mean profits, PL_i , will be negatively correlated with HC_i .

Means of HC_i for the different trader groups are reported in Table 4. Two statistics are of note. First, while the means for all five likely hedger types exceed 50%, none exceeds 60.3% so there are numerous cases in which individual traders in the likely hedger category hold positions opposite in sign to the aggregate likely hedger position. In these cases, the traders are either speculating or their hedging needs run counter to the majority of likely hedgers and we would expect them to actually benefit from any risk premium on these positions. Second, the mean HC_i for hedge funds is 22.1% and the median is only 18.6%. These are far lower than any other trader type. Clearly hedge funds generally take positions opposite to those of likely hedgers.

4.3. *Directional versus spread trading strategies*

Speculator trading strategies differ. Some may speculate that the price will rise or fall in the future by taking large long or short positions respectively (which we term “directional trades”); others may speculate on future price relationships by longing some contracts and simultaneously shorting others (“spread trades”). Examples of the latter would include: 1) calendar spreads in which the speculator longs some maturities and shorts others of the same contract type, and 2) crack spreads in which the speculator longs (shorts) crude oil and shorts (longs) gasoline and/or heating oil. We have a couple of goals in this area. First, anticipating that directional positions are much riskier than spread positions, we seek to quantify the risk difference.¹² Second, we seek to test whether and how average profits depend on this strategy choice. If, as anticipated, directional

¹² Note that because the denominator in equation 4 is total dollar open interest, $\%PL_{i,t}$ will tend to be smaller in absolute terms for traders taking spread positions. Since total open interest is the same for the following four positions: A) long 200 March 1995 crude oil contracts, B) long 100 March 1995 crude contracts and short 100 June 1995 crude contracts, C) long 100 crude, short 50 gasoline, and short 50 heating oil - all March 1995, D) long 125 March 1995 and short 75 March 1996 crude contracts, ignoring price differences, the denominator in equation 4 is the same for all four. However, the equation 4 numerator should normally be smaller in absolute terms for spread positions B and C than for trader A. Position D lies somewhere in between. Thus the time series standard deviation of $\%PL_{i,t}$ provides a measure of the riskiness of trader i 's trading strategy, i.e., the extent to which i hedges by taking spread positions.

trades are riskier than spread trades, average profits might have to be higher in order to attract directional speculators. On the other hand, it may be that market mispricings lead to arbitrage possibilities which astute spread traders can exploit leading to higher profits on spread trades. Büyükşahin et al. (2008) present evidence that prior to 2002 “near- and long-dated futures prices [for crude oil] were priced as though traded in separate markets” implying that during our data period there were price differences that calendar spread traders could profitably exploit.

Let $L_{i,c,m,t}$ represent trader i 's open interest position in contract c - m on day t if long and $S_{i,c,m,t}$ her position if short. In other words, $L_{i,c,m,t} = X_{i,c,m,t}$ if $X_{i,c,m,t} > 0$ and $= 0$ otherwise and $S_{i,c,m,t} = -X_{i,c,m,t}$ if $X_{i,c,m,t} < 0$ and $= 0$ otherwise. We measure how often trader i tends to take spread (versus directional) positions (SP) as:

$$SP_{i,t} = \frac{2 * \text{Min}[\sum_{c=1}^3 \sum_{m=1}^M L_{i,c,m,t}, \sum_{c=1}^3 \sum_{m=1}^M S_{i,c,m,t}]}{\sum_{c=1}^3 \sum_{m=1}^M (L_{i,c,m,t} + S_{i,c,m,t})} \quad (7)$$

If trader i holds equal numbers of long and contracts on day t , $SP_{i,t} = 1$. If i holds only long or only short positions, $SP_{i,t} = 0$. We calculate a summary measure for each trader i , i.e., SP_i by averaging $SP_{i,t}$ over all days t with non-zero open interest. Note that SP_i measures only the degree to which trader i 's *futures* position is hedged. If we included cash and forward market positions, then SP_i would tend to be close to one for hedgers but since cash and forward positions are not observed, a hedger's SP measure will be lower.

Means of SP_i by trader type are reported in Table 4 where the statistic that stands out is the mean of 15.0% for hedge funds. Their median SP is only 5.0%. Clearly the majority of hedge funds during this period were not trading calendar or crack spreads and were not hedging their futures positions by longing some contracts and simultaneously shorting others. Instead, they generally placed large directional bets by going all long or all short - somewhat more often all long according to the figures in Table 3. Moreover, as just seen with the HC measure, they were

generally longing when likely hedgers in the aggregate were short and shorting when likely hedgers in the aggregate were long.

In summary, we expect the standard deviation of trader i 's trading profits, $\%PL_{i,t}$, to be a negative function of SP_i . The risk premium and mispricing hypotheses imply different coefficient signs for the relation, if any, between i 's mean trading profits, $\%PL_i$, and SP_i .

4.4. Trader size

Ceteris paribus, we expect larger traders to be more informed than smaller traders. There are two reasons. First, spending funds to obtain information is more cost effective for large positions than small. Spending say \$100,000 to obtain information is harder to justify for a 100 contract position than for a 10,000 contract position. Second, larger energy firms could have superior information because they themselves are a large part of the energy market or observe a large part. Since the traders' identities are unknown to us, we can only measure size in terms of their open market positions so hypothesize that trading profits will tend to be higher for traders with large open market positions on average. As shown in Table 3, by this measure the largest traders are investment banks followed by commercial banks.

4.5. Trader activity

Along the same lines, we hypothesize that trading profits will be positively correlated with i 's trading activity. While this need not always be the case, we expect hedgers to tend to hold their positions longer than speculators. If they do, then the risk premium hypothesis would imply that traders who turn their positions over frequently should have higher pre-trading-cost profits on average. Since our trading profit calculations are before deducting trading costs, confirmation of this hypothesis does not necessarily imply higher profits after transaction costs. On the other hand, Odean (1999), and Barber and Odean (2000) find that more active household traders tend to lose money in the stock market even before deducting transaction costs. To test the relation between

trading activity and profits, we measure each trader i 's position turnover rate, $TURN_i$, by dividing i 's average daily trading volume by i 's average open interest position. The two averages are calculated only over days with non-zero open interest at either the beginning or end of the day. As reported in Table 4, turnover rates are highest for market makers/floor traders followed by hedge funds and individuals and lowest for commercial and investment banks.

In addition, it is possible that traders who make few trades are less skillful due to the lack of experience, implying lower trading profits. To test this, we relate trading profits to the percentage of the 962 days in our sample when the trader maintained a non-zero open interest position expecting a positive coefficient. Note that this test is subject to a possible survivorship bias because traders with losses may tend to drop out.

4.6. Long or short relative to the futures term structure

According to the risk premium hypothesis, hedging pressure will tend to push futures prices away from expected future spot prices, creating expected profit opportunities for speculators. Accordingly, in section 4.2, we hypothesized that trader i 's mean profits would be negatively correlated with our measure HC_i of the extent to which the sign of trader i 's open interest positions matched the aggregate positions of hedgers. However, HC_i may be an imperfect measure of this tendency since the likely hedger group on which HC_i is based undoubtedly includes some speculators. An alternative approach is to relate trader profits to the slope of the futures term structure. Futures markets may be in backwardation or contango because traders expect spot prices to fall or rise respectively. But they may also be in backwardation because hedgers are generally short (and hence push the futures price below the expected future spot price) or in contango because hedgers are generally long. If so, then speculators may be able to profit on average by longing the more distant contracts when futures market is in backwardation and shorting when in contango. We define a measure, BC , of the extent to which they follow this strategy. Specifically,

$$BC_i = \frac{\sum_{c=1}^C \sum_{m=1}^M \sum_{t=1}^T D_{i,c,m,t} |X_{i,c,m,t}|}{\sum_{c=1}^C \sum_{m=1}^M \sum_{t=1}^T |X_{i,c,m,t}|} \quad (8)$$

where, letting $m=n$ designate the nearby contract, if $mn > 1$, $D_{i,c,m,t} = 1$ if trader i 's open interest position in contract $c-m$ on day t is long when $P_{c,m,t} < P_{c,n,t}$, or short when $P_{c,m,t} > P_{c,n,t}$. If $m=n$, we define $D_{i,c,m,t} = 1$ if trader i 's open interest position in the nearby contract on day t is long when $P_{c,n,t} < P_{c,3,t}$, or short when $P_{c,n,t} > P_{c,3,t}$ where $P_{c,3,t}$ is the third month contract. If futures price patterns, i.e., backwardation versus contango, are partially due to hedging pressure, then trader profits should be positively correlated with BC_i . Means of BC by trader classification are reported in the final column of Table 4. Except possibly for the commercial bank category for which the mean of BC_i is 39.2%, differences between different trader classifications are small. Unlike the other measures in Table 4, the null that BC does not differ by trader classification cannot be rejected at the .05 level.

5. Results

5.1. Trading profits and luck.

First, we test Hartzmark's (1991) contention that futures trading profits are determined solely by luck. For this, we test whether the number of traders tending to make consistent profits or losses is more than one would expect due to chance by estimating an analysis of variance (ANOVA) based on the 183,634 daily percentage profit observations, $\%PL_{i,t}$. One issue in an ANOVA estimation is whether to assume variances are the same or different. As explained above, we expect profit variances to differ by trader since profits should be less variable for traders taking spread positions than for those whose open interest positions are mostly long or mostly short. Testing this using the Brown-Forsythe test for homogeneity of variances, with an F stat of 99.98, the homogeneity null is rejected at the .0001 level. Given this result, we employ Welch's ANOVA

statistic which presumes differing variances. With an F of 2.15, the null that mean profit differences among the 382 traders are due to chance is rejected at the .0001 level.

For further evidence on this issue, for each of our 382 traders, we calculate z-values for the test of the null hypothesis that their mean daily trading profits, i.e., the mean of $\%PL_{i,t}$ across all days t , is zero. By chance approximately 3.82 traders should have mean profits significantly greater than zero at the 1% level. In our sample, 12 traders do. The expected number with profits significantly greater than zero at the 5% level is 19.1 while 35 actually are. At the other end of the spectrum, 22 have mean profits significantly less than zero at the 1% level and 58 at the 5% level. To test whether trader profit/loss patterns are persistent, we calculate $\%PL_i$ over both the June 1993 - May 1995 and June 1995 - March 1997 sub-periods restricting the sample to the 224 traders with at least 25 trades and 50 observations in each sub-period. The correlation coefficient of $+.140$ is significantly different from zero at the .05 level. Clearly some traders tend to make persistent profits and even more persistent losses.

5.2. Trading profits and trader type

Next we test for mean profit differences between our likely hedger (first five categories in Table 1) and likely speculator categories (categories 6-8 in Table 1). Results are presented in Panel A of Table 5. In columns 2-5, we present statistics based on the individual trader/day observations, $\%PL_{i,t}$. Also, for each day t , we combine the open interest positions of all likely hedgers into a combined hedger position and calculate the percentage profits on this combined position. We do the same for likely speculators. Means and standard deviations for these combined positions are in columns 6 and 7 respectively. Hence the means in column 3 are unweighted treating each $\%PL_{i,t}$ equally while the means in column 6 are weighted by the size of each trader's open interest position that day. For the statistics in columns 2-5, there are 97,510 hedger/day observations and 50,561 speculator/day observations; for the stats in columns 6-7, there are 962 day observations for both.

Having seen that profit variances differ substantially by trader, the p-values in column 4 are based on standardized profits $\%PL_{i,t}/\sigma_i$ where σ_i is the time series standard deviation of $\%PL_{i,t}$.

For likely hedgers, the average unweighted daily trading profit is -0.0154% (-3.80% annualized). For likely speculators, it is +0.0394% (10.40% annualized). Both are significantly different from zero and each other at the .0001 level. The average combined or weighted daily profit is -.0173% (-4.27% annualized) for likely hedgers and .0312% (8.18% annualized) for likely speculators. Due to the reduced number of observations, significance levels are lower for the combined position. Still the two combined means are both significantly different from zero and each other at the .05 level. In summary, the mean profits of likely hedgers and speculators are consistent with the risk premium hypothesis.

In panel B, the same statistics are presented for the eleven trader type classifications. Consistent with the risk premium hypothesis, unweighted mean profits are negative for all five likely hedger classifications and positive for all three likely speculator classifications, though two of the combined means switch signs. However, within the likely hedger category, mean profits are significantly less than zero¹³ (at the 1% level for the unweighted means and 5% level for the combined means) only for refiners and commercial banks. Within the likely speculator categories, mean profits are significantly positive only for hedge funds/money managers for which they are a sizable 15.2% of open interest when annualized. Since we were unsure whether the bulk of the trades by investment banking houses would be hedges or speculations, it is interesting that their mean profits are small and insignificantly different from zero, suggesting a mix. The profit figures in Table 5 are prior to transaction costs. When we calculate profits deducting round-trip transaction

¹³ While this statement is for a one-tailed test, the p-values in Table 4 are two-tailed since different signs are expected for different categories and no sign was hypothesized for three.

costs of \$15 per contract¹⁴ mean daily profit figures are reduced by .003 to .005 percentage points but remain significantly positive for both hedge funds and speculators in general.

Interestingly, the trader category with the second largest mean loss is market makers. Their unweighted daily mean of $-.0370\%$ implies an annualized loss of -8.9% .¹⁵ This is consistent with the hypothesis that market makers tend to lose money on their overnight inventory positions due to trades with more informed investors (Bagehot, 1971; Glosten and Milgrom, 1985). It is also consistent with prior studies showing that information advantages possessed by market makers are short-lived at best (Silber, 1984; Hasbrouck and Sofianos, 1993; Yao, 1997).¹⁶ However our data does not allow us to rule out the possibility that market makers possess information advantages in intra-day trading that they acquire by observing order flow (Manaster and Mann, 1996; Brown and Zhang, 1997; and Ready, 1999). Indeed, it is unlikely that market makers/floor traders would continue in this line of work if these overnight losses were representative of their overall trading profits. Still it is clear that, at least on positions held overnight, market makers/floor traders have no special skills or knowledge and tend to lose on their trades with more informed investors, consistent with the aforementioned literature.

In Panel A of Table 6, we report hedger/speculator distributions of those traders whose mean daily trading profits are significantly greater or less than zero at the .05 level. While the overall sample is approximately evenly split between likely hedgers and speculators, over two-thirds of the traders with significant losses are likely hedgers and almost three-quarters of those with significant trading profits are likely speculators. The null that the distribution is the same is

¹⁴ Estimates for transaction costs for small traders in energy futures markets revolve around this figure. Haigh and Holt (2002) use roundtrip costs of \$15 per contract while Girma and Paulson (1998) use \$100 for the roundtrip cost of six contracts. This figure likely overstates transaction costs for the large traders in our sample.

¹⁵ Generous estimates of the bid-ask spread would lower this daily loss figure only .003 to .005 percentage points in absolute terms so it is still large and significant.

¹⁶ It is also possible that these overnight losses could represent a disposition effect as documented by Odean (1998a) even among these supposedly sophisticated traders.

rejected at the .01 level. In Panel B of Table 6, we report the distribution across all eleven trader types of the traders with significantly positive or negative profits at the 5% level. The two categories that stand out here are hedge funds and market makers. While hedge funds make up 18.8% of the sample of 382 traders, they account for 40% of those with significant positive profits and only 3.4% of those with significant negative trading profits. Conversely market makers account for 18.1% of the total sample but 36.2% of those with significant negative profits.

In summary, we find significant differences in futures trading profits by trader type with likely speculators tending to make profits and likely hedgers losses. Hedge funds are particularly profitable. Market makers tend to make substantial losses on positions held overnight.

5.3. Risk premia or superior information?

While the mean trading profits of likely hedgers and speculators are consistent with the risk premium hypothesis, a possible alternative explanation for the profit difference is that speculators have superior information or skills so make profits at the expense of hedgers who are less informed or skilled. Furthermore, if one regards hedge funds as particularly likely to have an information or skill advantage, the finding that their profits are higher than other speculators may be viewed as consistent with the information hypothesis. Next we explore whether profit patterns are more consistent with the risk premium or information hypotheses.

According to the risk premium hypothesis, if hedgers are long (short) on balance, futures prices will be pushed above (below) expected future spot prices, creating positive expected profits for traders who short (long). So first we investigate whether futures prices reflect hedging pressure. Directly testing whether hedging pressure pushes futures prices away from the expected future spot price is complicated by the difficulty in measuring expected future spot prices. However, since futures prices converge to the spot price as maturity approaches, a further implication is that, ceteris paribus, futures prices will tend to fall (rise) over time when hedgers on balance are long (short). To test this, we measure net hedging pressure in contract c - m on day t , $NHP_{c,m,t}$ as the

aggregate long positions of likely hedgers minus aggregate likely hedger short positions divided by total open interest in contract c-m on day t. The hedging pressure hypothesis implies that $\% \Delta P_{c,m,t} = (P_{c,m,t} - P_{c,m,t-1}) / P_{c,m,t-1}$ will be negatively correlated with $NHP_{c,m,t-1}$. To pick up the impact of $NHP_{c,m,t-1}$ on $\% \Delta P_{c,m,t}$ it is important to control for other forces causing changes in $P_{c,m,t}$, especially changes in the expected future spot price. Expecting these forces to have similar impacts across the futures term structure, we approximate them with the percentage change in the price of the nearby futures contract. Letting m=n designate the nearby contract, $\% \Delta P_{c,n,t}$ represents the percentage change in the nearby contract. We regress $\% \Delta P_{c,m,t}$, where $m > n$, on $\% \Delta P_{c,n,t}$ and $NHP_{c,m,t-1}$. To ensure accurate measures of net hedging pressure we restrict the sample to contracts with at least five traders on day t-1. Results of this regression are reported in Table 7. The coefficient of $NHP_{c,m,t-1}$ is negative and significantly different from zero at the .01 level implying that when hedgers in the aggregate are net long (short), the futures price is pushed above (below) the expected future spot price potentially creating profit opportunities for speculators.

Next we explore whether the profit differences between likely hedgers and likely speculators are, in fact, due to the latter taking advantage of the potential profit opportunities created by hedging pressure on futures prices. In section 4.2. and equation 6, we defined a “hedger concordance” measure, HC_i , of the extent to which the signs (long or short) of trader i’s open interest positions match the signs of the aggregate open interest positions of likely hedgers. If trader i always longs when likely hedgers are net long and shorts when likely hedgers are net short, $HC_i = 1$. If i always shorts when likely hedgers are net long and longs when likely hedgers are net short, $HC_i = 0$. If traders profit from the impact of net hedging pressure on futures prices by taking positions opposite to those of hedgers, then trader profits should be negatively correlated with HC_i .

In Model 1 in Table 8, we report the regression of trader i’s average daily percentage trading profits, $\% PL_i$ on HC_i estimated over the 382 traders i. Since we observed above that some trader’s daily profits are more variable than others, t-values are corrected for heteroskedasticity using the White procedure. With a t-value of -7.686, trader profits are a strong negative function of

HC_i . The parameter estimates in Model 1 imply that annualized percentage trading profits are a surprising 4563 basis points higher for a trader who always longs (shorts) those contracts in which the majority of likely hedgers are short (long) versus a trader whose long and short positions always match the signs of the aggregate positions of likely hedgers. Of course there are no traders who are always in agreement or disagreement with the net aggregate hedger positions. To be more realistic, suppose we compare a trader whose long/short position agrees with the aggregate likely hedger position 75% of the time with one whose position disagrees 75% of the time, i.e., $HC_i = .75$ versus $HC_i = .25$. Approximately 25.9% of the traders in our sample are outside these two bounds and most hedge funds HC_i 's are below .25. According to the parameter estimates in Model 1 of Table 8, annualized annual trading profits tend to be approximately 2266 basis points higher for the trader with $HC_i = .25$. This is a sizable difference which is considerably more than the speculator/hedger profit difference documented in Table 5.

To determine whether the profit differences between likely hedgers and likely speculators documented in Table 5 are due to speculators exploiting the risk premium by taking positions opposite to likely hedgers, we add HC_i to a regression with trader type dummy variables. First however, Model 2 in Table 8 presents the results of a regression of $\%PL_i$ on zero-one dummy variables for likely hedgers, speculators, market makers, and investment banks without HC_i . Unclassified traders are the left out group. Hence, the coefficients measure the profit difference between that trader type and the unclassified group so are not particularly important in themselves. More important is the difference between the coefficients of the likely hedger and speculator dummies which is negative and implies that the annualized trading profits of likely speculators are approximately 1420 basis points higher than those of likely hedgers. When we run an F-test based on the difference in the error sum of squares between Model 2 and a version of Model 2 in which equal hedger and speculator coefficients are imposed, the null that there is no difference, *ceteris paribus*, between hedger and speculator profits is rejected at the .001 level.

In Model 3, we add the variable HC_i to this regression. HC_i is again highly significant. Importantly, the likely hedger and speculator coefficients are not significantly different from zero or each other. Indeed the likely hedger coefficient is the larger of the two. While it is possible that some individual speculators profit because of unique information or skills, the Model 3 results imply that the higher profits of speculators as a group are due to their exploitation of the risk premia caused by hedging pressure - not systematic information differences.

Next we estimate regressions with dummies for all trader types except the unclassified group with and without the HC_i variable. We are particularly interested in whether hedge fund profits are due to exploitation of the risk premium since, as seen in Table 5, they are the most profitable of the eleven trader types and, as seen in Table 4, they tend to take positions opposite to the aggregate hedger position more often than any other trader type. Estimation results with trader type dummies only are reported in Model 4 in Table 8 and with HC_i included in Model 5. The results in Model 4 largely repeat those in Table 5 indicating profit differences between trader types with hedge funds are the most profitable trader type. When we run an F-test based on the difference in the error sum of squares between Model 4 and a version of Model 4 in which equal coefficients are imposed for all trader types save market makers, the null that profits do not differ by trader type is rejected at the .001 level.¹⁷ However, when HC_i is added to the regression in Model 5, the hedge fund coefficient is no longer significant. Moreover, after controlling for HC_i , the null that profitability does not differ among trader types (excluding market makers) cannot be rejected at even the .10 level. In summary, excluding market makers, differences in mean profitability by trader type (and hedge fund profits in particular) appear solely due to differences in the extent to which they take advantage of the risk premium by taking positions opposite to the net

¹⁷ To test whether this result was solely due to the high hedge fund coefficient in Model 4, we also test whether the differences between the remaining eight trader type coefficients are significant using a F-test based on the difference in the error sum of squares between regressions with equal and unequal coefficients for all types except market makers and hedge funds. The null that, ceteris paribus, profits do not differ between the eight trader types is rejected at the .01 level.

hedger position. This implies that the profits of speculators in general, and hedge funds in particular, represent compensation for the services they provide hedgers by accepting risk and providing liquidity - not superior information or skill .

A final issue is whether trading profit differences among individual traders within the hedger and speculator categories are related to HC_i . As discussed above, the figures in Table 3 suggest that some traders within the likely hedger category may be speculating or employing a selective hedging strategy. Also hedging needs differ so some hedgers may hedge by longing (shorting) when the majority are short (long). In columns 2 and 3 of Table 9, we regress $\%PL_i$ on HC_i for the hedger subset. Again the coefficient of HC_i is negative and significant indicating that profit differences among likely hedgers are a function of the extent to which the sign (long or short) of their open interest positions matches or differs from the majority. In the last two columns of Table 9, we regress $\%PL_i$ on HC_i for the speculator subset. Again the coefficient of HC_i is negative and significant indicating that speculators who take positions opposite in sign to the net aggregate hedger positions tend to make higher profits than those following other strategies. In other words, those speculators who follow a strategy of longing (shorting) when a majority of hedgers are short (long) make higher profits than those who do not.

5.4. Determinants of trading profit volatility

We observed above that some traders' trading profits vary much more day-to-day than others and hypothesized that this was because some traders tend to employ spread trades with roughly equal long and short positions so that their overall futures positions are roughly delta neutral while others take mostly short or mostly long positions. The latter could either be speculators betting on whether energy prices will rise or fall or hedgers whose overall positions are hedged but whose futures positions are not. To test whether profit volatility differences are due to differences in spreading activity, in Table 10, we regress the time-series standard deviation of trader i 's daily trading profits (as a percent of her total open interest) on SP_i , our measure of the extent to

which i 's futures positions are hedged by balancing longs and shorts as defined in equation 7. We anticipate a negative coefficient, i.e., lower volatility on spread positions. We also include the log of trader i 's average open interest position anticipating that larger traders may tend to hedge their positions. The regression is estimated first without trader type dummy variables and then with (with unclassified traders as the omitted category).

As expected, the volatility of a trader's profits is largely determined by whether she tends to take spread positions. According to SP_i 's Model 1 coefficient, the standard deviation of a trader's percentage profits is about 1.03 percentage points higher if they always take either solely long or short positions versus if they always hold equal numbers of long and short energy contracts. Since the average standard deviation is only 1.09%, this is a considerable difference.

5.5. Other determinants of trading profitability

In addition to trader type and HC_i , we hypothesized above that trader profits would covary with trader size, number of days in the market, portfolio turnover, whether the trader generally takes spread or directional positions as measured by SP_i , and whether the trader longs or shorts when the market is in backwardation or contango, as measured by BC_i . In Table 11, trader i 's mean daily percentage profits/losses, $\%PL_i$, is regressed on these measures along with zero-one dummies for the ten trader types excluding unclassified traders. Results are shown in Model 1 in Table 11. In Model 2 we add an interaction variable representing the turnover rate for household traders only and in Model 3 drop all trader type dummies except market makers. We discuss results for each of the determinants hypothesized in section 4 in turn.

5.5.1. Hedger concordance

The hedger concordance variable, HC_i , remains highly significant when these other variables are added to the equation and its coefficient actually increases (in absolute terms) from that estimated in Table 8. Again we find that, consistent with the risk premium hypothesis, trader

profits tend to be considerably higher if the trader generally longs (shorts) when a majority of hedgers are short (long). This is the most important determinant of individual trader profits.

5.5.2. Spread versus directional positions

We anticipated that profitability could differ between traders taking all long or all short positions and traders taking spread positions. As noted above, Büyükşahin et al. (2008) present evidence indicating profitable calendar spread opportunities prior to 2002. It is possible that crack spread opportunities were also present. On the other hand, profits could be higher on totally long or short positions due to their greater riskiness as documented in Table 10. The positive coefficient for SP_i in Table 11 is consistent with the former hypothesis but is only significant at the .05 level in Model 3. In Table 4 we observed that hedge funds generally take overwhelmingly long or overwhelmingly short positions - not spread positions. Since the coefficient of SP_i is positive in the Table 11 regressions, it is clear that the high hedge fund profits are not due to this strategy. Instead hedge fund profits are explained by their exploitation of the risk premium.

5.5.3 Long or short relative to the futures term structure

In section 4.6., we hypothesized that to the extent the slope of the futures price term structure is due to hedging pressure, not just expectations of future spot prices, expected trading profits to a strategy of longing distant contracts when the market is in backwardation and shorting in contango would be positive and defined a measure, BC_i , of the extent to which trader i pursued this strategy. The risk premium hypothesis implies a positive coefficient for BC_i . As reported in Table 11, the coefficient of BC_i is positive as hypothesized and significant at the .01 level.

5.5.4. Trader size

We hypothesized in section 4 that larger traders would tend to be more informed so would tend to have higher profits, *ceteris paribus*. However, the coefficient of the size variable is

insignificant and is actually negative. Clearly there is no evidence larger traders have an informational advantage.

5.5.5. *Trading activity*

Profits are a positive, but insignificant, function of the percentage of the 962 days that the trader held a non-zero open interest position. However, as expected, mean trader profits vary positively with turnover, $TURN_i$, indicating that traders who turn their positions over more often tend to have higher profits - at least before transaction costs. Since this seemingly contradicts the finding of Odean (1999), and Barber and Odean (2000) that in the stock market more active household traders tend to make losses, it is possible that the relationship between turnover and profitability is different for households. To test this, we add an interaction variable, IND_TURN_i , equal to $TURN_i$ when i is a household and zero otherwise. Results are shown in Model 2 in Table 11. Clearly, the turnover-profit relation is different for household traders since the coefficient of IND_TURN_i is negative and significant. The sum of the two coefficients is an insignificant $-.0108$ indicating little relation between turnover and profits for households.

5.5.6. *Trader type*

As in Table 9, the results in Table 11 show that market makers tend to lose money on positions held overnight after controlling for other determinants of trading profits, indicating that any informational advantages they possess are short-lived and that they tend to lose money on trades with more informed investors. Aside from the difference between investment banks and unclassified traders, there are no significant profit differences between any of the other trader type pairs and the trader type dummies as a group are insignificant excepting market makers. When we drop all trader type dummies except the market maker dummy in Model 3, the adjusted R^2 actually improves. In summary, except for market makers, all other profit differences between trader types are explained by our independent variables - the HC_i variable in particular.

5.6. *Robustness Checks*

Since a majority of hedgers tend to be short in these markets and since there was a slight increase in energy prices over our data period (Figure 1), some have questioned whether our results (for HC_i in particular) might be picking up a tendency to profit by longing futures. To test this, we added a variable measuring the percent of trader i 's open interest positions that were long (the remainder being short) to Model 3 in Table 11. This variable was insignificant (p -value = .34) and the other coefficients were little changed from those reported in Table 11.

As documented in Table 10 and earlier, some traders profits are much more variable than others. Accordingly we use White's heteroskedasticity correction in all standard error estimations. As a robustness check, we also estimated Model 3 in Table 11 with trader i 's standardized profits, i.e., mean/standard deviation, as the dependent variable. Due to the change in scale, the coefficients differ and adjusted R^2 falls slightly to .301 but all significant variables in Table 11 remain significant and both insignificant variables remain insignificant.

5.7. *Hedgers and speculators*

Finally, in Table 12, we estimate separate Model 3 relations for likely speculators and hedgers. One difference stands out. In the likely speculator regression, the coefficient of the turnover measure is large and strongly significant for all except households indicating that hedge funds and energy traders who turn over their positions quickly tend to have higher profits than those who turn their positions over more slowly. For hedgers, this variable is insignificant.

6. **Conclusions**

We find that approximately 35% of the variation in mean profits among different individual large traders in energy futures markets can be explained by differences in their trading objectives (hedging versus speculation) and strategies. We find strong support for the risk premium hypothesis. First, mean speculator profits are significantly higher than mean hedger profits.

Second, our evidence indicates that hedging pressure pushes futures prices away from expected future spot prices. Third, individual traders (whether hedgers or speculators) who generally short when likely hedgers in the aggregate are net long and long when likely hedgers in the aggregate are net short make considerably higher profits on average than traders who generally long (short) when likely hedgers are long (short). Fourth, traders who long when the market is in backwardation and short when in contango make higher profits than those who do the opposite.

Classifying futures traders into eleven line-of-business types, we find significant mean profit differences with hedge funds the most profitable. However, excepting market makers, mean profit differences among the other ten trader types are completely explained by the extent to which they exploit the risk premium by longing (shorting) when hedgers are short (long). Hedge funds in particular tend to exploit the risk premium and to earn higher than normal profits due to this strategy. We further find that, excepting households, speculators with higher turnover rates tend to have higher profits. Market makers tend to have large and significant trading losses on the positions that they hold overnight indicating that they tend to lose on their trades with more informed investors and that any informational advantage they may have from observing the order flow is short-lived.

While it is possible that some individual traders profit from superior information or skills, we find no evidence of systematic information differences in our data. In particular, there is no evidence of systematic information or skill differences between hedgers and speculators or among the ten trader types since their profit differences are completely explained by the risk premium, i.e., by whether they tend to long (short) when hedgers in the aggregate are long (short) or the reverse. Also, there is no evidence that larger traders profit at the expense of smaller traders.

In summary, we find that the main determinant of a trader's mean profits is whether she contributes to the risk premium by longing (shorting) when hedgers in the aggregate are net long (short) or exploits the risk premium by shorting (longing) when hedgers in the aggregate are net long (short). In other words, our evidence indicates that the profits of speculators in general, and

hedge funds in particular, are due to the liquidity and/or risk absorption services they provide hedgers by being willing to short (long) when hedgers in the aggregate are net long (short).

REFERENCES

- Ackermann, Carl, Richard McEnally, and David Ravenscraft, 1999, The performance of hedge funds: Risk, return, and incentives, *Journal of Finance* 54, 833-874.
- Adam, Tim R. and Chitru S. Fernando, 2006, Hedging, speculation, and shareholder value, *Journal of Financial Economics* 81, 283-309.
- Bagehot, Walter, 1971, The only game in town, *Financial Analysts Journal* 27, 12-14 & 22.
- Barber, Brad and Terrance Odean, 2000, Trading is hazardous to your wealth: The common stock investment performance of individual investors, *Journal of Finance* 55(2), 773-806.
- Bessembinder, Hendrik, 1992, Systematic risk, hedging pressure, and risk premiums in futures markets, *Review of Financial Studies* 5, 637-667.
- Bessembinder, Hendrik, and Kalok Chan, 1992, Time varying risk premia and forecastable returns in futures markets, *Journal of Financial Economics* 32(2), 169-93.
- Bodnar, Gordon M., Gregory S. Hayt and Richard C. Marston, 1998 Wharton survey of derivatives usage by U.S. non-financial Firms, *Financial Management* 27(4), 70-91.
- Brown, David P. and Zhi Ming Zhang, 1997, Market orders and market efficiency, *Journal of Finance*, 52, 277-308
- Brown, Gregory W., Peter R. Crabb, and David Haushalter, 2006, Are firms successful at selective hedging? *Journal of Business* 79, 2925-2949.
- Bryant, Henry L. and Michael S. Haigh, 2004, Bid-ask spreads in commodity futures markets, *Applied Financial Economics* 14, 923-936
- Büyükhahin, Bahattin, Michael Haigh, Jeffrey Harris, James Overdahl, and Michel Robe, 2008, Fundamentals, trader activity, and derivative pricing, working paper, Commodity Futures Trading Commission.
- Büyükhahin, Bahattin, and Jeffrey H. Harris, 2009, The role of speculators in the crude oil futures market, working paper, Commodity Futures Trading Commission.
- Carhart, Mark, 1997, On persistence in mutual fund performance, *Journal of Finance* 52, 57-82.
- Chan, Hsiu-Lang, Narasimhan Jegadeesh, and Russ Wermers, 2000, The value of active mutual fund management: An examination of the stockholdings and trades of fund managers, *Journal of Financial and Quantitative Analysis* 35, 343-368.

De Roon, Frans, Theo Nijman, and Chris Veld, 2000, Hedging pressure effects in futures markets, *Journal of Finance* 55, 1437-1456.

Dolde, Walter, 1993, The trajectory of corporate financial risk management, *Journal of Applied Corporate Finance* 6, 33-41.

Ederington, Louis and Jae Ha Lee, 2002, Who trades futures and how: Evidence from the heating oil futures market, *Journal of Business* 75, 353-373.

Fama, Eugene, and Ken French, 1987, Commodity futures prices: Some evidence on forecast power, premiums, and the theory of storage, *Journal of Business* 60, 55-73.

Girma, Paul B. and Albert S. Paulson, 1998, Seasonality in petroleum futures spreads, *Journal of Futures Markets* 18, 581-598.

Glaum, Martin, 2002, The determinants of selective exchange risk management – evidence from german non-financial corporations, *Journal of Applied Corporate Finance* 14, 108-121.

Glosten, Lawrence R. and Paul R. Milgrom, 1985, Bid, ask, and transaction prices in a specialist market with heterogeneously informed traders, *Journal of Financial Economics* 14, 71-100.

Grinblatt, Mark, and Sheridan Titman, 1989, Mutual fund performance: An analysis of quarterly portfolio holdings, *Journal of Business* 62, 393-416.

Grinblatt, Mark, and Sheridan Titman, 1993, Performance measurement without benchmarks: An examination of mutual fund returns, *Journal of Business* 66, 47-68.

Haigh, Michael S. and Matthew T. Holt, 2002, Crack spread hedging: Accounting for time-varying volatility spillovers in the energy futures markets, *Journal of Applied Econometrics* 17, 269-289.

Haigh, Michael S., Jana Hranaiova, and James A. Overdahl, 2007, Price volatility, liquidity provision, and the role of hedge funds in energy futures markets, *Journal of Alternative Investments* 4, 10-38.

Hartzmark, Michael, 1987, Returns to individual traders of futures: Aggregate results, *Journal of Political Economy* 95, 1292-1306.

Hartzmark, Michael, 1991, Luck versus forecast ability: Determinants of trader performance in futures markets, *Journal of Business* 64, 49-74.

Hasbrouck, Joel, and George Sofianos, 1993, The trades of market makers: An empirical analysis of NYSE specialists, *Journal of Finance* 48, 1565-1593.

Hicks, John R., 1939, *Value and Capital* (Oxford University Press, Cambridge)

Ibbotson, Roger, and Peng Chen, 2005, Sources of hedge fund returns: Alphas, betas, and costs, Working paper 05-17, International Center for Finance, Yale University.

Interagency Task Force on Commodity Markets, 2008, Interim Report on Crude Oil, US Commodity Futures and Trading Commission, Washington, DC.

Jagannathan, Ravi, Alexey Malakhov, and Dmitry Novikov, 2008, Do hot hands persist among hedge fund managers? An empirical evaluation, Working paper on SSRN.

Keynes, John M., 1930, A Treatise on Money, Vol II (MacMillan, London).

Kolb, Robert, 1992, Is normal backwardation normal? Journal of Futures Markets 1(12), 75-91.

Kosowski, Robert, Narayan Naik, and Melvyn Teo, 2007, Do hedge funds deliver alpha? A bayesian and bootstrap analysis, Journal of Financial Economics 84, 229-264.

Leuthold, Raymond, Philip Garcia, and Richard Lu, 1994, The returns and forecasting ability of large traders in the frozen pork bellies futures market, Journal of Business 67(4), 459-473.

Manaster, Steven and Steven C. Mann, 1996, Life in the pits: Competitive market making and inventory control, Review of Financial Studies 9, 953-975.

Odean, Terrance, 1998a, Are investors reluctant to realize their losses? Journal of Finance 53, 1775 - 1798.

Odean, Terrance, 1998b, Volume, volatility, price, and profit when all traders are above average, Journal of Finance 53, 1887-1934.

Odean, Terrance, 1999, Do investors trade too much?, American Economic Review 89, 1279-98.

Ready, Mark J., 1999, The specialist's discretion: Stopped orders and price improvement, Review of Financial Studies 12, 1075-1112.

Silber, William L., 1984, Marketmaker behavior in an auction market: An analysis of scalpers in futures markets, Journal of Finance 34, 937-953.

Stulz, René, 1996, Rethinking risk management, Journal of Applied Corporate Finance 19, 8-24.

US Senate Staff Report, 2006, The role of speculation in rising oil and gas prices: A need to put the cop back on the beat, Staff Report of the Permanent Subcommittee on Investigations, Senate Print: 109-65.

Wang, Changyun, 2001, Investor sentiment and return predictability in agricultural futures markets, Journal of Futures Markets 21(10), 929-52.

Wermers, Russ, 2000, Mutual fund performance: An empirical decomposition into stock-picking talent, style, transaction costs, and expenses, *Journal of Finance* 55, 1655-94.

Yao, Jian, 1997, Spread components and dealer profits in the interbank foreign exchange market, Working Paper.

Figure 1 - Crude, Heating and Gasoline Oil Prices June 1993 - March 1997

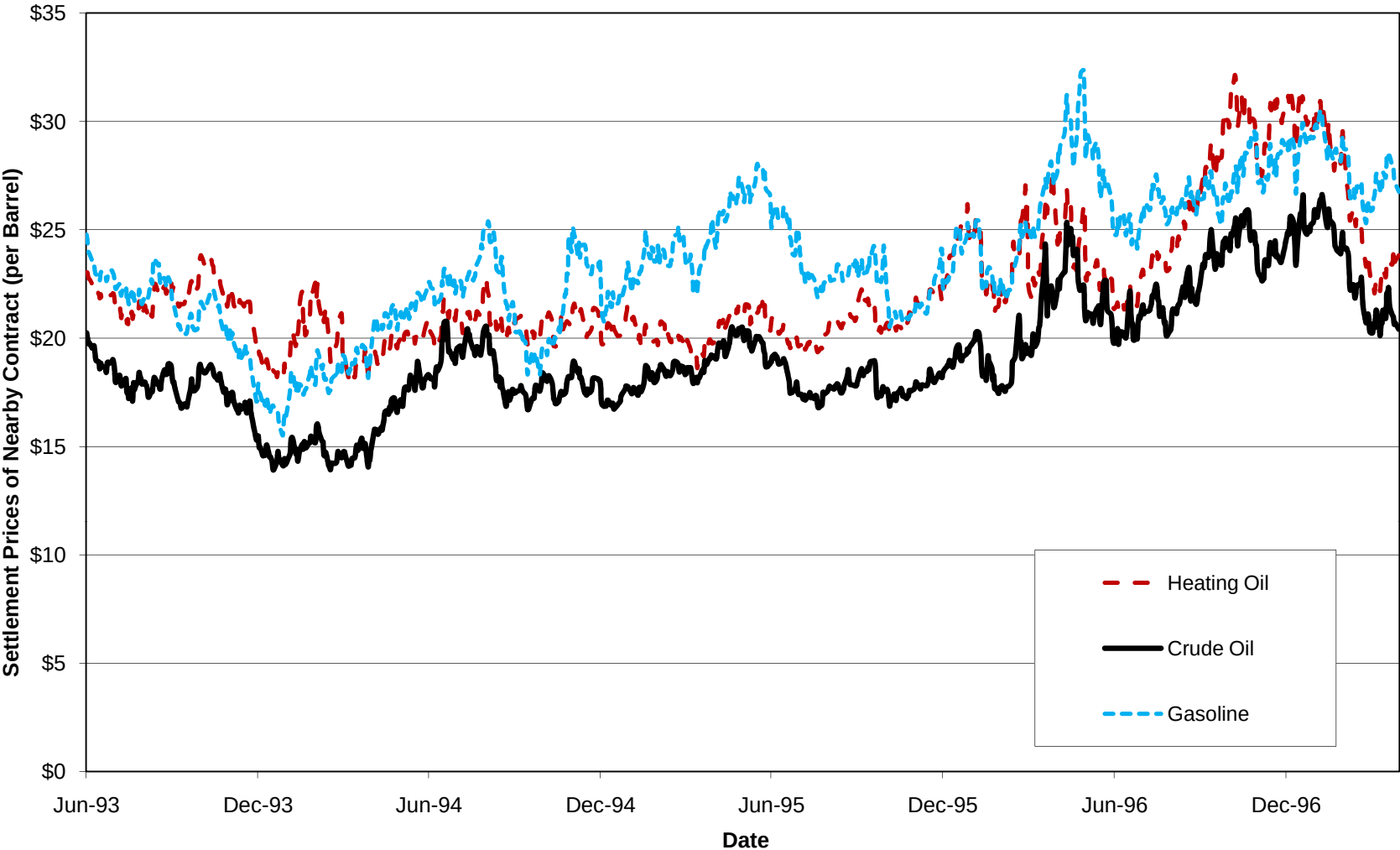


Figure 2 - Backwardation and Contango
Ratio of the Price of the Third Month Contract to the Nearby

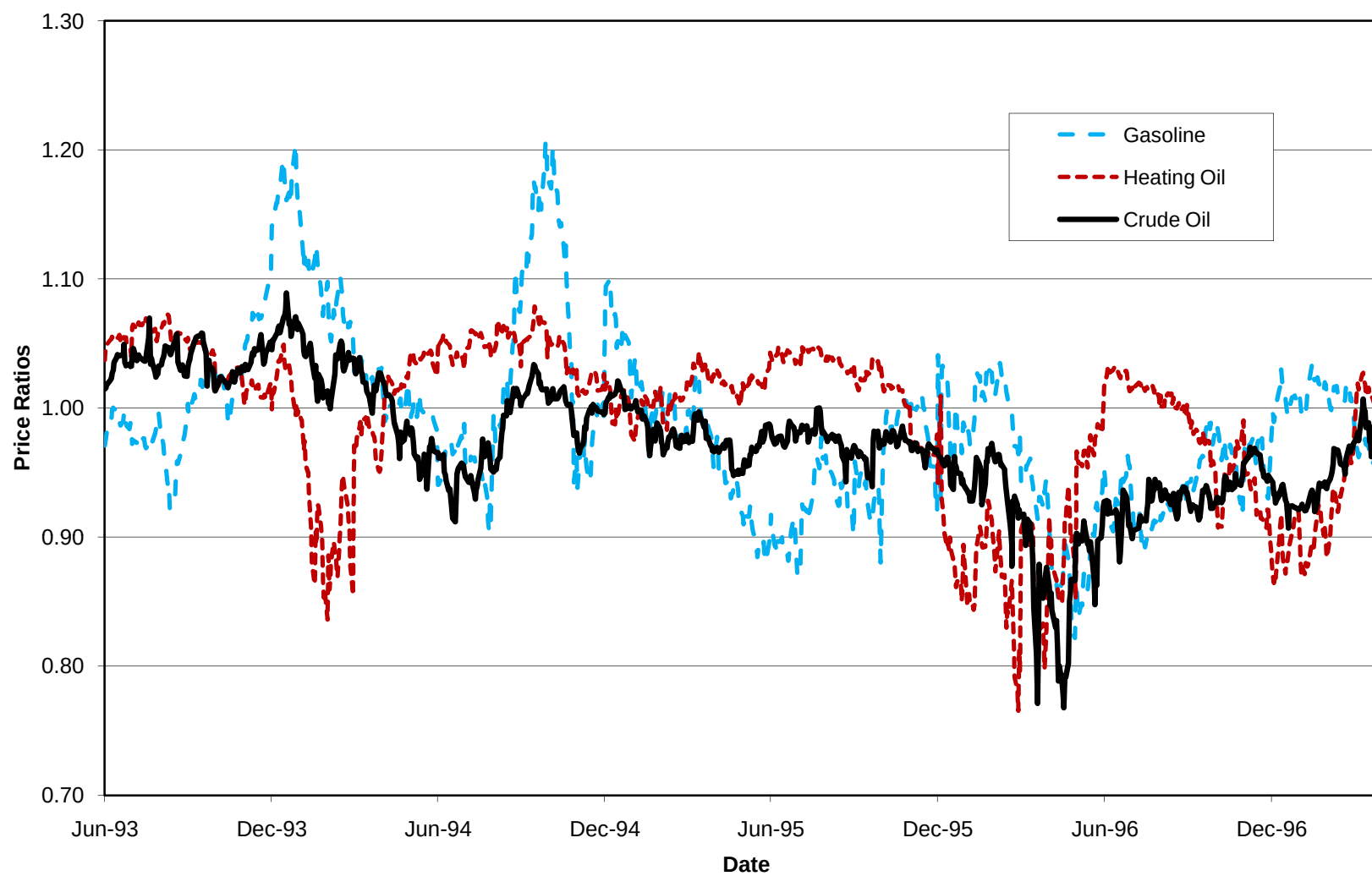


Table 1 - Trader type classifications

The energy futures trader classifications used in this study are described. Traders identified as “commercial” traders in the CFTC’s Large Trader Reporting System (LTRS) files were categorized into classifications 1-6 and 9 with the assistance of the Office of Policy at the Department of Energy. Classifications 7, 8, 10, and 11 are based on the CFTC’s classifications of “non-commercial” traders.

Classification		Description and classification procedure
Likely hedgers		
1	Refiners	Oil and gas firms with refineries.
2	Independent producers	Oil and gas producers without refinery capacity - commonly referred to as exploration and production companies.
3	Marketers/Distributors/ Pipelines	Pipeline companies and companies primarily involved in marketing and distributing oil and refined products. Some may produce oil as well but their primary business is in transporting, marketing, and/or distributing.
4	Large consumers	Non-energy companies which are consumers of refined oil products, e.g., airlines, trucking firms.
5	Commercial banks	Two digit SIC code: 60.
Likely Speculators		
6	Energy traders	Energy trading companies (mostly offshore) with no known energy assets classified as commercial by CFTC.
7	Hedge funds/ Money managers	Non-commercials identified in the CFTC’s LTRS files as commodity pool operators, commodity trading advisers, or money managers.
8	Households/Individuals	The residual category of non-commercial traders in the CFTC’s LTRS files.
Other		
9	Investment banks and brokers	Two-digit SIC code: 62.
10	Market makers /Floor traders	Exchange members trading for their own accounts. Includes trades for their own accounts by floor brokers and futures commission merchants.
11	Unclassified	Unidentified traders.

Table 2 - Energy futures trading data

Descriptive statistics are presented for large traders from the CFTC's LTRS files in the crude oil, gasoline, and heating oil futures markets for the June 1993 - March 1997 period. Statistics for the full data set of 939 traders are presented in Panel A and for our working data set of 382 more active traders in Panels B and C. Open interest figures are gross, i.e., longs + shorts.

	Crude oil	Gasoline	Heating oil	All three markets
Panel A - Full data set - 939 traders				
Avg. daily gross open interest (contracts)	607,085	125,160	192,619	924,864
Avg. daily gross open interest (\$ billion)	\$10,704	\$2,868	\$4,234	\$17,806
Trader/contract/day observations	538,097	211,215	310,304	1,059,616
Panel B - Working data set - 382 traders				
Avg. daily open interest (contracts)	593,723	123,165	189,522	906,410
Avg. daily open interest (\$ billion)	\$10,456	\$2,821	\$4,162	\$17,439
Trader/contract/day observations	516,172	203,864	300,438	1,020,474
Trades	228,532	118,705	139,097	486,334
Avg daily open int. per trader (contracts)	4,404	1,243	1,748	4,579
Avg trade size (contracts)	264	158	151	206
Panel C - Time-to-expiration of open interest positions				
Mean time-to-expiration (months)	14.6	1.6	2.8	5.0
Median time-to-expiration (months)	5.8	1.6	2.3	2.7
Cumulative percent of contracts expiring in				
1 month or less	14%	32%	22%	18%
2 months or less	27%	62%	45%	41%
3 months or less	35%	78%	60%	54%
6 months or less	50%	97%	88%	73%
12 months or less	62%	99%	99%	88%
24 months or less	72%	100%	100%	96%

Table 3 - Descriptive statistics

Descriptive statistics are presented for the eleven trader line-of-business categories described in Table 1. The percentages of the 1,020,474 trader/day/contract observations accounted for by each category are reported in column 3 (% obs). The percentages of total open interest accounted for by each are reported in column 5 and the same for each of the three markets in columns 8-10. In the last three columns (11-13), we report the percentage of the open interest positions in each market which are long (the remainder being short).

Trader categories	# traders	% of total obs	% of total trades	% of total open int.	Mean daily trader O.I. (contracts)	Mean trade size in contracts	% of market open int.			% Long (of open interest positions)		
							Crude oil	Gasoline	Heat. oil	Crude oil	Gasoline	Heat. oil
Refiners	57	28.1%	28.1%	28.6%	5,842	213	26.0%	36.5%	31.5%	45.2%	31.4%	34.9%
Independent producers	14	2.4%	1.4%	2.2%	2,690	160	2.6%	0.5%	1.8%	30.1%	68.8%	26.8%
MDPs	22	12.2%	12.1%	12.8%	7,665	210	10.4%	17.9%	17.2%	54.3%	42.5%	48.5%
Energy consumers	6	0.7%	0.7%	0.7%	2,197	211	0.5%	0.2%	1.9%	76.7%	42.5%	85.5%
Commercial banks	16	9.3%	9.1%	9.1%	8,950	159	11.6%	3.2%	4.9%	37.4%	85.6%	70.8%
Energy traders	14	5.8%	4.4%	4.5%	4,574	151	5.1%	3.1%	3.4%	54.2%	47.8%	35.7%
Hedge funds	72	7.7%	7.7%	6.9%	1,916	289	5.7%	7.9%	10.1%	63.4%	65.4%	54.1%
Households/Individuals	28	2.6%	3.0%	1.8%	1,612	190	2.1%	1.5%	1.0%	66.2%	87.0%	71.3%
Investment banks	19	17.9%	17.5%	27.3%	21,606	285	29.9%	21.3%	23.1%	55.6%	73.9%	48.2%
Market makers	69	9.4%	15.0%	4.7%	1,687	132	5.0%	4.3%	4.0%	42.8%	49.5%	58.8%
Unclassified	65	3.9%	3.8%	1.4%	576	114	1.0%	3.6%	1.1%	41.4%	28.0%	42.9%

Table 4 - Trader type characteristics

Means of hypothesized determinants of trader profitability are reported. Hedger confluence, HC, is a measure of the extent to which the signs (long versus short) of a trader's positions in various future contracts tend to match the signs of the aggregate net positions of likely hedgers in the same contracts on the same days as defined in equation 6 in the text. The spread measure, SP, is a measure of the extent to which the trader holds spread positions as defined in equation 7 in the text. The percentage of days in the market reports the percentage of the 962 days in our sample when the trader's open interest position was not zero. Turnover is measured as mean daily trading volume divided by mean total open interest (long positions plus short positions). BC is a measure of the extent to which a trader longs distant contracts when the market is in backwardation and shorts when in contango as defined in equation 8.

	Hedger confluence (HC)	Spread measure (SP)	Percentage of days in market	Turnover (TURN)	Backwardation Contango (BC)
Refiners	60.3%	43.9%	79.4%	13.3%	52.0%
Independent producers	59.5%	38.3%	54.5%	9.0%	53.5%
MDPs	57.6%	51.1%	69.5%	15.9%	53.2%
Large consumers	51.3%	20.5%	54.3%	11.0%	50.1%
Commercial banks	57.4%	38.6%	58.4%	7.9%	39.2%
Energy traders	50.3%	44.6%	64.8%	13.3%	54.3%
Hedge funds	22.1%	15.0%	48.6%	22.4%	48.5%
Households/Individuals	40.5%	32.7%	39.6%	21.9%	51.4%
Investment banks	49.7%	46.9%	60.9%	8.3%	49.5%
Market makers	40.9%	44.4%	42.6%	35.3%	49.8%
Unclassified	58.0%	33.2%	37.6%	18.8%	55.4%

Table 5 - Trader profits

Statistics are presented for traders' daily trading profits before transaction costs as a percentage of the trader's total open interest. Statistics are presented for likely hedgers and likely speculators in Panel A and for eleven separate trader type categories in Panel B. The p-values are for tests of the null that the mean percentage profits are zero. Statistics are based on trades and open interest positions of 382 large traders. * and ** denote coefficients significantly different from zero at the .05 and .01 levels respectively.

Panel A - Likely hedgers and speculators

Category	Individual trader/day observations				Combined positions - 962 day obs.		
	Obs.	Mean	p-value	Std. Dev.	Mean	p-value	Std. Dev.
Likely hedger	97510	-0.0154%**	.0001**	1.0548%	-0.0173%	.0212*	0.2330%
Likely speculator	50561	0.0394%**	.0001**	1.2288%	0.0312%	.0329*	0.4540%
Panel B - Trader type categories							
Refiners	42760	-0.0129%**	.0087**	.9808%	-0.0193%	.0476*	0.3026%
Independent producers	7077	-0.0204%	.1567	1.0859%	-0.0407%	.0640	0.6819%
MDPs	14594	-0.0062%	.5903	0.8461%	0.0035%	.4550	0.1440%
Large consumers	2999	-0.0095%	.6466	1.3516%	-0.0230%	.5050	1.0684%
Commercial banks	8838	-0.0397%**	.0001**	.8371%	-0.0252%	.0218*	0.3409%
Energy traders	8557	0.0113%	.6283	0.9165%	-0.0030%	.6573	0.2132%
Hedge funds	32011	0.0561%	.0001**	1.3295%	0.0519%	.0415*	0.7901%
Households/Individuals	9993	0.0102%	.2298	1.1548%	0.0119%	.5556	0.6259%
Investment banks	11037	0.0021%	.7146	.7671%	0.0066%	.3719	0.2306%
Market makers	24537	-0.0370%	.0001**	0.9346%	-0.0228%	.0014**	0.2210%
Unclassified	21242	-0.0159%	.1248	1.3176%	-0.0086%	.5697	0.4665%

Table 6 - Distribution of highly profitable and unprofitable traders

Percentages by trader type are reported for: a) traders whose mean daily trading profits are significantly positive at the .05 level, b) traders with significantly negative mean daily trading profits at the .05 level, and c) the full sample. Hedger/speculator proportions are reported in Panel A; proportions by trader type in Panel B. In panel A the sample is restricted to those identified as likely hedgers or speculators.

Trader category	Sample		
	Profits significantly positive at 5% level	Profits significantly negative at 5% level	Overall
Panel A - Likely hedgers and speculators - sample restricted to likely hedgers or speculators			
Likely hedgers	25.9%	68.0%	50.2%
Likely speculators	74.1%	32.0%	49.8%
Panel B- Eleventrader categories - full sample			
Refiners	11.4%	13.8%	14.9%
Independent producers	2.9%	2.9%	3.7%
MDPs	5.7%	1.7%	5.8%
Large consumers	0.0%	0.0%	1.6%
Commercial banks	0.0%	10.3%	4.2%
Energy traders	5.7%	1.7%	3.7%
Hedge funds	40.0%	3.4%	18.8%
Households/Individuals	11.4%	8.6%	7.3%
Investment banks	2.9%	3.4%	5.0%
Market makers	11.4%	36.2%	18.1%
Unclassified	8.6%	17.2%	17.0%

Table 7 - Hedging pressure and futures prices

The percentage change in the price of futures contract c-m from day t-1 to t, $\% \Delta P_{c,m,t}$, is regressed on the percentage change in the price of the nearby futures contract, $\% \Delta P_{c,n,t}$, and a measure of net hedging pressure, $NHP_{c,m,t-1}$. $NHP_{c,m,t-1}$ is measured as the aggregate long positions of likely hedgers in contract c-m on day t-1 minus aggregate short positions of likely hedgers in that contract that day divided by total open interest in contract c-m on day t. The sample is restricted to contracts with maturities m exceeding the nearby contract and to days with at least 5 traders with open interest positions in contract c-m. The reported p-values are for rejection of the null that the coefficient is zero.

	Coefficient	p-value
Intercept (x100)	.0050	.2302
$\% \Delta P_{c,n,t}$	.4739	.0001
$NHP_{c,m,t-1}$ (x 100)	-.0546	.0065
Adjusted R ²	.595	
Observations	34224	

Table 8 - Trader profits and risk premia exploitation

Trader i 's mean daily profits/losses (as a percent of her mean total open interest position) over the data period are regressed on: 1) dummies for the trader type, and 2) a measure, HC_i , of the extent to which the signs (long versus short) of trader i 's positions match the aggregate positions of likely hedgers.

$$HC_i = \sum_{c=1}^C \sum_{m=1}^M \sum_{t=1}^T D_{i,c,m,t} |X_{i,c,m,t}| / \sum_{c=1}^C \sum_{m=1}^M \sum_{t=1}^T |X_{i,c,m,t}| \text{ where } X_{i,c,m,t} \text{ is trader } i\text{'s open interest}$$

position (long or short) in contract c - m on day t . $D_{i,c,m,t} = 1$ if $X_{i,c,m,t}$ is long when likely hedgers are net long in the aggregate or short when likely hedgers are net short in the aggregate. $D_{i,c,m,t} = 0$ if $X_{i,c,m,t}$ is short when likely hedgers are net long in the aggregate or long when likely hedgers are net short in the aggregate. t -values with the White correction for heteroskedasticity are reported in parentheses. The regression is estimated over 382 traders. * and ** denote coefficients significantly different from zero at the .05 and .01 levels respectively.

	Model 1	Model 2	Model 3	Model 4	Model 5
Intercept	.0799 ** (6.691)	-.0233 (-1.637)	.0952** (4.404)	-.0233 (-1.623)	.0971 (3.881)
Hedger concordance (HC_i)	-.1853** (-7.686)		-.2043** (-6.830)		-.2077** (-5.622)
Likely hedger		.0074 (0.481)	.0091 (0.633)		
Likely speculator		.0599** (3.487)	.0030 (0.160)		
Market maker		-.0233 (-1.238)	-.0582** (-3.186)	-.0233 (-1.228)	-.0588** (-3.124)
Investment bank		.0261 (1.286)	.0092 (0.534)	.0261 (1.276)	.0089 (0.511)
Refiner				.0125 (0.773)	.0173 (1.139)
Independent producer				-.0031 (-0.104)	.0002 (0.007)
MDP				0.0128 (0.799)	.0121 (0.781)
Large consumer				.0381 (1.580)	.0241 (0.942)
Commercial bank				-.0207 (-0.986)	-.0219 (-1.092)
Energy trader				.0390* (1.990)	.0232 (1.223)
Hedge fund				.0766** (3.931)	.0021 (0.082)
Household/Individual				.0275 (1.227)	-.0088 (-0.429)
Adjusted R^2	.148	.088	.202	.096	.198

Table 9 - Separate hedger and speculator regressions

Trader i's mean daily profits/losses (as a percent of her mean total open interest position) over the data period are regressed on a measure, HC_i , of the extent to which the sign (long versus short) of trader i's positions match the aggregate positions of likely hedgers.

$$HC_i = \sum_{c=1}^C \sum_{m=1}^M \sum_{t=1}^T D_{i,c,m,t} |X_{i,c,m,t}| / \sum_{c=1}^C \sum_{m=1}^M \sum_{t=1}^T |X_{i,c,m,t}| \text{ where } X_{i,c,m,t} \text{ is trader } i\text{'s open interest}$$

position (long or short) in contract c-m on day t. $D_{i,c,m,t} = 1$ if $X_{i,c,m,t}$ is long when likely hedgers are net long in the aggregate or short when likely hedgers are net short in the aggregate. $D_{i,c,m,t} = 0$ otherwise. Equations are estimated separately for the likely hedger and likely speculator sub-samples. t-values with the White correction for heteroskedasticity are reported in parentheses. * and ** denote coefficients significantly different from zero at the .05 and .01 levels respectively.

	Likely hedger subsample		Likely speculator subsample	
	Coefficient	t-value	Coefficient	t-value
Intercept	.0907	2.397*	.0784	5.584**
Hedger concordance (HC_i)	-.1812	-2.686**	-.1287	-3.445**
Adjusted R^2	.095		.058	
Observations	115		114	

Table 10 - Determinants of futures trading profit volatility

The standard deviation of trader i 's daily percentage profits/losses is regressed on: 1) a measure (defined in equation 7) of the extent to which the trader hedges her futures position by taking spread positions, SP_i , in which she balances long and short positions, 2) trader i 's size as measured by the mean log of her total open interest. If i always balances long positions with an equal number of short positions, $SP_i=1$. If she always takes either all long or all short positions, $SP_i=0$. In model 2, unclassified commercial traders are the left out category. The regression is estimated over 382 traders. * and ** denote coefficients significantly different from zero at the .05 and .01 levels respectively.

Variable	Model 1		Model 2	
	Coefficient	t-value	Coefficient	t-value
Intercept	1.9391	37.842**	1.9789	32.377**
Spread position measure, SP_i	-1.0293	-32.075**	-1.0104	-29.395**
Log of average open interest	-.0707	-9.363**	-.0740	-7.560**
Refiner			.0277	.796
Independent producer			.0201	.401
MDP			-.0245	-.558
Large consumer			.1610	2.265*
Commercial bank			-.1156	-2.286*
Energy trader			-.0208	-.409
Hedge fund			-.0206	-.706
Households/Individual			-.0977	-2.587*
Investment bank			-.0495	-.987
Market maker			-.552	-1.871*
Adjusted R^2	.810		.819	

Table 11 - Futures trading profitability regressions

Trader i 's mean daily profits/losses, as a percent of her mean total open interest position, are regressed on: 1) a measure, HC_i (defined in equation 6) of the extent to which the signs (long versus short) of trader i 's positions match the aggregate positions of likely hedgers, 2) a measure, SP_i , (defined in equation 7) of the extent to which i takes spread positions, 3) a measure, BC_i of how trader i 's positions relate to whether the market is in backwardation or contango, 4) trader i 's size as measured by the mean log of her total open interest, 5) i 's turnover, $TURN_i$, measured as average daily trading volume divided by average daily open interest, 6) the percentage of days the trader was in the market, 7) an interaction variable, IND_TURN_i , equal to $TURN_i$ when i is a household and zero otherwise, and 8) zero-one dummies for the trader type. Unclassified traders are the left out trader category in Models 1 and 2. The regressions are estimated over 382 traders. * and ** denote coefficients significantly different from zero at the .05 and .01 levels respectively. t-values are based on White heteroskedasticity consistent standard errors.

Variable	Model 1		Model 2		Model 3	
	Coeff.	t-value	Coeff.	t-value	Coeff.	t-value
Intercept	.0252	0.493	.0214	0.429	-.0047	-0.105
Hedger concordance (HC_i)	-.2393	-7.420**	-.2408	-7.374**	-.2204	-10.117**
Spread trader measure (SP_i)	.0293	1.763	.0307	1.873	.0356	2.221*
Term structure slope (BC_i)	.1495	4.768**	.1488	4.723**	.1421	4.721**
Log of mean open interest	-.0059	-1.162	-.0058	-1.155	-.0013	-0.275
Turnover, $TURN_i$	.1633	3.050**	.1844	3.537**	.1767	3.572**
Percent of days in market	.0183	1.114	.0188	1.149	.0208	1.370
Refiners	.0252	1.801	.0291	1.820		
Independent producers	.0192	0.748	.0208	0.809		
MDP	.0154	1.046	.0152	1.034		
Large consumers	.0468	1.778	.0481	1.814		
Commercial bank	.0244	1.323	.0259	1.406		
Energy trader	.0292	1.768	.0296	1.796		
Hedge fund	-.0009	-0.047	-.0024	-0.115		
Households/Individual	-.0133	-0.596	.0282	1.007		
Investment bank	.0368	2.106*	.0381	2.177*		
Market maker	-.0889	-4.776**	-.0931	-5.023**	-.1013	-7.341**
IND_TURN_i			-.1952	-2.299*	-.1484	2.203*
Adjusted R^2	.340		.351		.354	

Table 12 - Hedger and speculator regressions

Model 3 in Table 11 is estimated separately for likely hedgers and likely speculators. * and ** denote coefficients significantly different from zero at the .05 and .01 levels respectively. t-values are based on White heteroskedasticity consistent standard errors.

Variable	Likely hedgers		Likely speculators	
	Coeff.	t-value	Coeff.	t-value
Intercept	.0640	0.993	-.1012	-1.319
Hedger concordance (HC_i)	-.2000	-3.555**	-.1647	-4.181**
Spread trader measure (SP_i)	.0260	1.109	-.0176	-0.644
Term structure slope (BC_i)	.1532	3.606**	.1605	2.756**
Log of mean open interest	-.0081	-1.260	.0086	0.776
Turnover, $TURN_i$	.0634	0.716	.3051	9.417**
Percent of days in market	.0079	0.302	-.0040	-0.109
IND_TURN_i			-.2056	-2.481*
Adjusted R^2	.270		.413	